# Agentic AI Recap Single and Multi-Agent Systems

From Passive Tools to Proactive Partners

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# Learning Objectives

1. Differentiate Agentic AI from previous paradigms
2. Define AI agents & key characteristics (autonomy, reactivity, proactiveness)
3. Master reasoning vs planning (Sense-Think-Act framework)
4. Understand & apply Chain-of-Thought, ReAct, Tree-of-Thoughts
5. Compare classical vs modern LLM-based agent architectures (including BDI mapping)
6. Explore Multi-Agent Systems: challenges, patterns & practical design

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# Agenda : Quick Tour

1. Background & Evolution of Agentic AI (4–5)
2. Core Agent Definitions & Concepts (6–7)
3. PEAS & Environment Classification (8–11)
4. Reasoning vs Planning Foundations (12)
5. Reasoning Techniques: CoT, ReAct, RAG (13–15)
6. Advanced Single-Agent Reasoning Patterns (16–17)
7. Multi-Agent Systems: Motivation & Challenges (18–19)
8. Multi-Agent System Patterns & Design (20–22)
9. Summary & Q&A (23–24)


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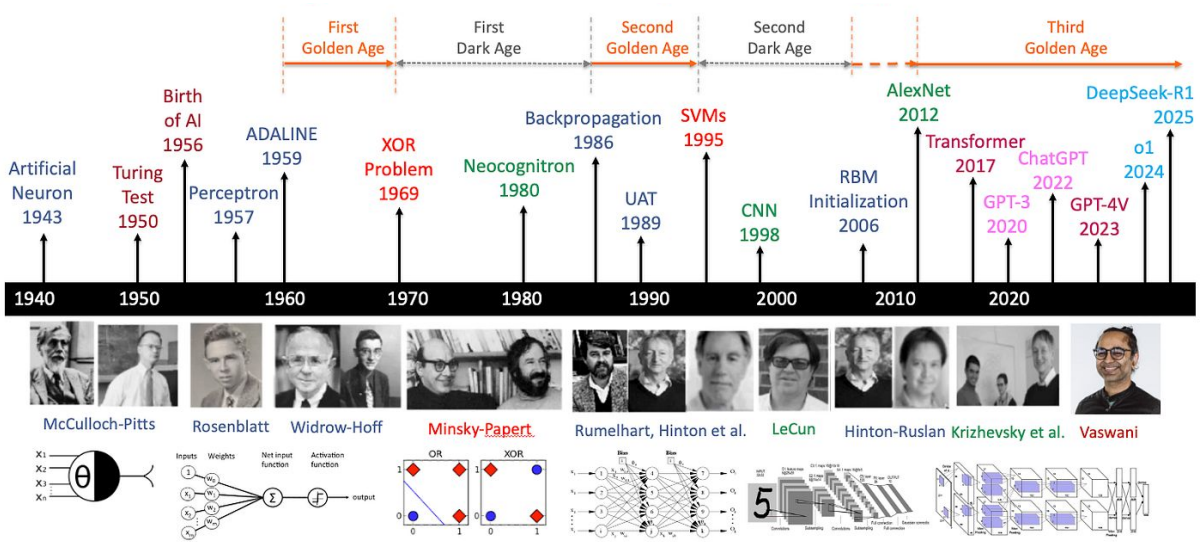
# Evolution of AI : The Quest for Intelligence

***1950s–80s*** *Symbolic AI: Rule-based systems*

***1990s–2010s*** *Machine Learning: Statistical pattern learning*

***2017– Present*** *Foundation Models: Scale + emergent reasoning*

***Now*** *Agentic AI: Goal-directed, autonomous action*




IMG: https://medium.com/@lmpo/a-brief-history-of-ai-with-deep-learning-26f7948bc87b



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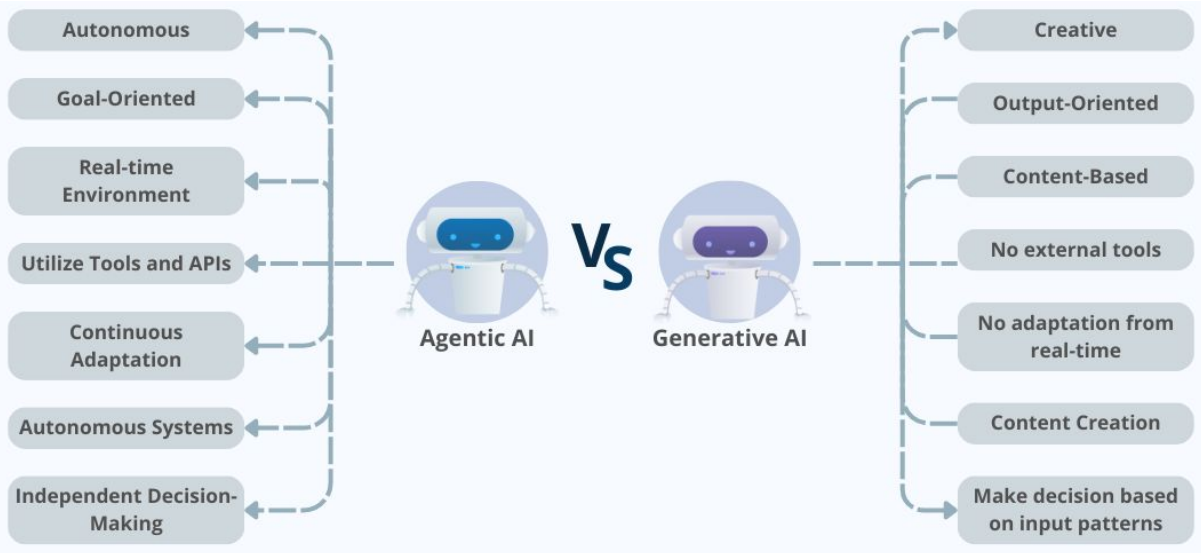
# From Passive Tools to Proactive Partners

A journey to understand the fundamental shift from AI as a **passive tool** towards AI as a **proactive partner** that can understand goals, make plans, and take actions in the world.




IMG: https://www.arabsolutionsgroup.com/2025/04/11/understanding-the-shift-from-generative-ai-to-agentic-ai/



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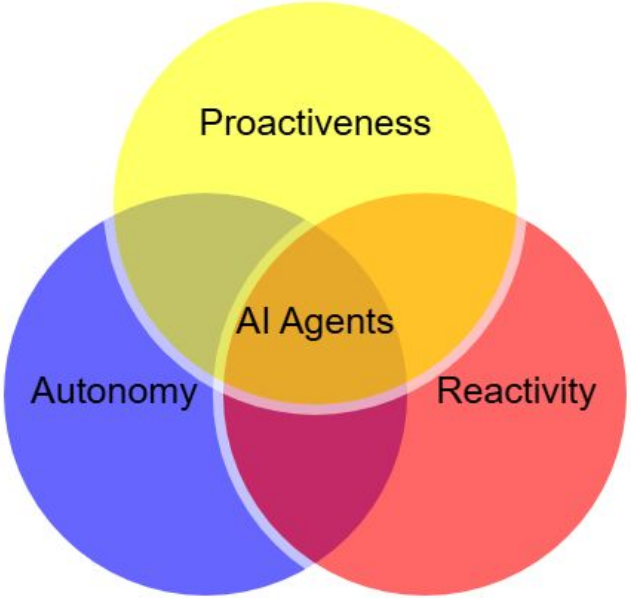
# What is an AI Agent?

An AI agent is a system that exhibits:

- **Autonomy:** Acts based on experience/adaptation (not just hard-coded rules)
- **Reactivity:** Perceives & responds to changes in environment
- **Proactiveness:** Takes initiative to achieve goals

**Example:** Robot Vacuum Cleaner after furniture is rearranged, briefly struggles, updates its map, avoids new obstacles, and resumes efficient cleaning on its own.





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# Perceived Agency : Building Human Trust

1. Humans naturally **attribute intentionality**, **free will**, even **consciousness** to agents
2. **Perceived Agency** significantly increases trust & willingness to delegate
3. **Caution**: Too much anthropomorphism can backfire (unrealistic expectations)

**Key takeaway:** Perceived agency is a design choice where balance is critical.

IMG: https://uwaterloo.ca/cerc-human-centred-robotics-machine-intelligence/towards-physical-social-human-robot-interaction



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# PEAS Framework

**P** – Performance measure (external success)
**E** – Environment
**A** – Actuators
**S** – Sensors

PEAS for Robot Vacuum Cleaner:

| Component | Example |
|---|---|
| Performance | Cleanliness / dirt removed |
| Environment | Rooms, furniture, dirt |
| Actuators | Wheels, suction, brush |
| Sensors | Dirt detector, bump sensors |


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# PEAS Examples

| Agent Type | Performance Measure | Environment | Actuators | Sensors |
|---|---|---|---|---|
| Chess / Go player | win / lose / draw | game board | moves | board position |
| Medical diagnosis system | accuracy of diagnosis | patient, staff | display questions, diagnoses | keyboard entry of symptoms |
| Part-picking robot | percentage of parts in correct bins | conveyor belt with parts, bins | jointed arm and hand | camera, joint angle sensors |
| Refinery controller | purity, yield, safety | refinery, operators | valves, pumps, heaters, displays | temperature, pressure, chemical sensors |
| Interactive English tutor | student's score on test | set of students, testing environment | display exercises, suggestions, corrections | keyboard entry |


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# Environment Classification

| Dimension | Easy End | Hard End | Examples |
|---|---|---|---|
| Observability | Fully observable | Partially observable | Chess (full) vs. Poker (partial – hidden cards) |
| Determinism | Deterministic | Stochastic | Sudoku vs. Driving in traffic |
| Dynamics | Static | Dynamic | Crossword puzzle vs. Taxi driving (world changes while deciding) |
| Agents | Single-agent | Multi-agent | Solitaire vs. Chess (competitive) or collaborative robotics |


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# Increasing levels of agent capability:

1. **Simple Reflex Agents:** React directly to current percepts
2. **Model-Based Agents:** Maintain an internal model of the world
3. **Goal-Based Agents:** Choose actions to achieve future goals
4. **Utility-Based Agents:** Select actions that maximize preferences (utility)

**Key Idea:** Each level adds flexibility and enables more intelligent behavior.


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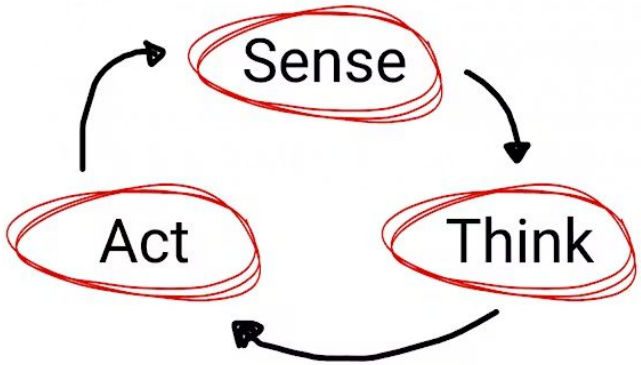
# Reasoning vs Planning : Core Distinction

- **Reasoning:** Local, immediate next step
  ("What should I do right now?")

- **Planning:** Global, future-oriented
  ("What sequence gets me to the goal?")




**Sense–Think–Act cycle:**
Sense → **Think** (reason locally + plan globally) → Act

IMG: https://www.corporate-rebels.com/blog/sense-think-act-repeat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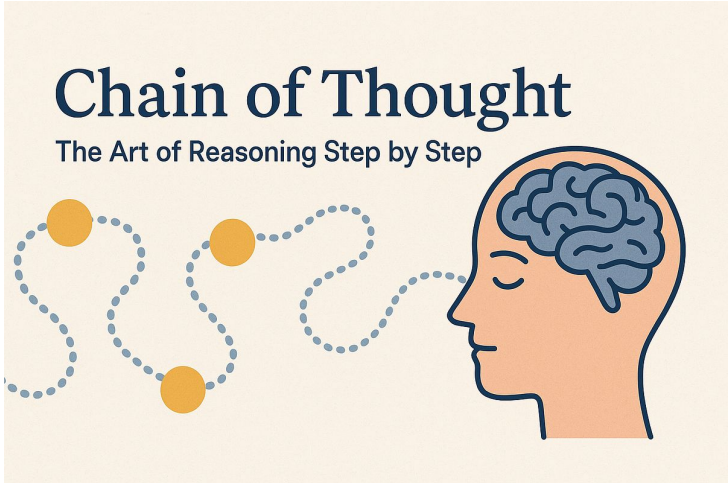
# Chain-of-Thought (CoT) : The Foundation

Most important prompt technique of the last few years:
**"Let's think step by step."**

**Variants:**

1. Zero-shot CoT
2. Few-shot CoT (with examples)
3. Least-to-Most prompting

**Limitation:** Purely internal — no tool use




IMG: https://blog.stackademic.com/chain-of-thought-the-art-of-reasoning-step-by-step-1e254640b270?gi=6f179692aeef



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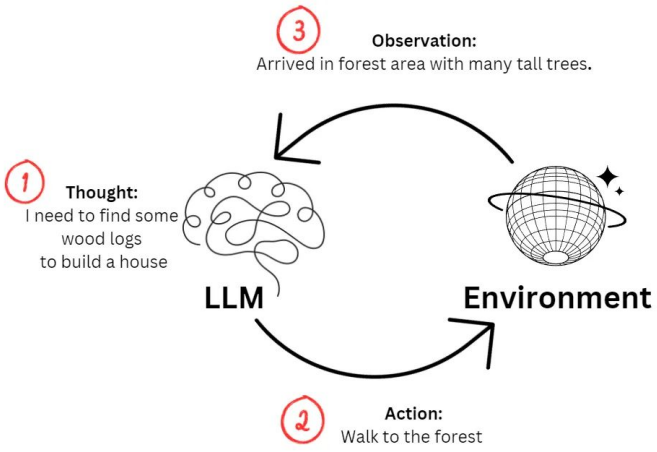
# ReAct : Reasoning + Acting Loop

**ReAct = Reason + Act**

Thought → Action → Observation → Thought → … until done




## Advantages over Plan-and-Execute:

- Adaptive (can change plan based on new info)
- Recovers from mistakes
- Uses tools naturally

IMG:https://asycd.medium.com/react-prompt-framework-enhancing-ais-decision-making-with-human-like-reasoning-72a30df34ead



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# RAG + Agents : The Knowledge Triad

**Modern agents usually combine:**

1. Knowledge Base (documents, databases)
2. Retriever (vector search)
3. Agentic Generator (ReAct loop with tools)

Very few serious agents today work purely from parametric memory. Almost all use retrieval.

IMG:https://yourgpt.ai/blog/general/rag-chatbot-vs-ai-agent


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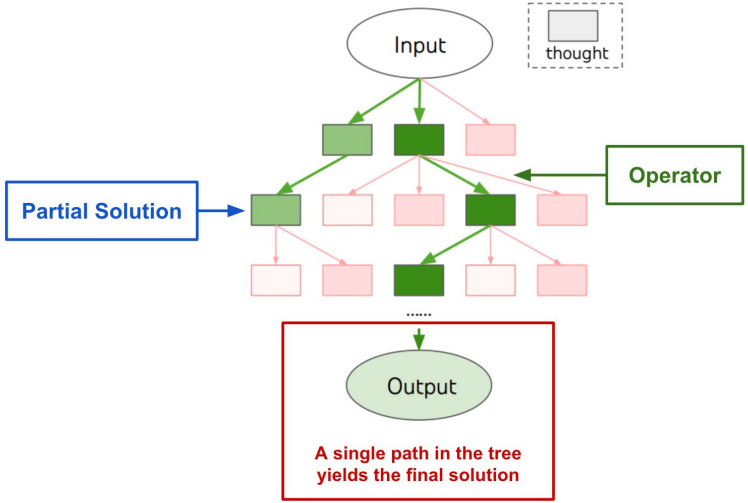
# Tree of Thoughts (ToT) : Branching Exploration

**Advanced reasoning pattern:**

1. Generate multiple candidate thoughts
2. Evaluate each
3. Prune bad paths / expand promising ones
4. Repeat




**When to use:**
Complex multi-step problems where linear CoT fails

IMG:https://cameronrwolfe.substack.com/p/tree-of-thoughts-prompting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# Self-Critique & AI Judges

**Self-Critique loop:**

Generate → Critique own output → Refine → Repeat

**AI Judges:**

Use another LLM to score/choose best outputs

- Scalable alternative to human evaluation
- Must be calibrated against human judgments

IMG:https://towardsdatascience.com/llm-as-a-judge-a-practical-guide/



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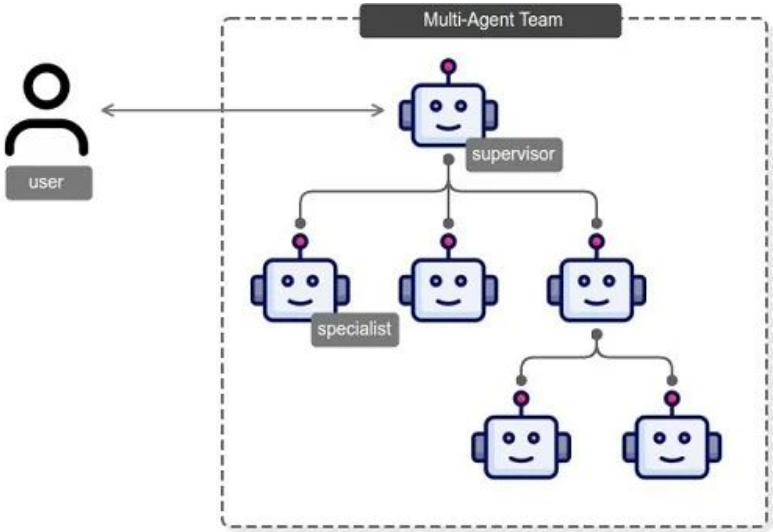
# Multi-Agent Systems : Why & When

MAS = Multiple agents that **coordinate, communicate, collaborate**




**Add more agents when you need:**

- Higher **quality** through diverse perspectives
- Lower **latency** through parallelism
- **Specialization** (writer, critic, researcher, planner…)

IMG:https://simplified.com/blog/automation/multi-ai-agent-collaboration



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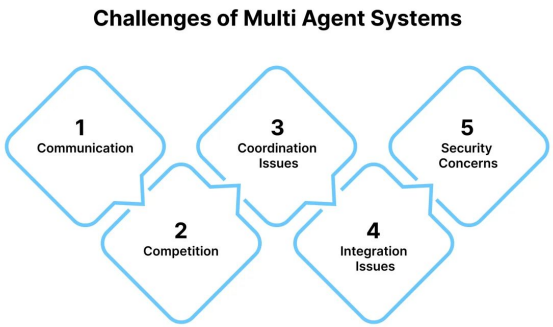
# MAS Challenges (The Hard Parts)

- **Coordination:** Who does what? When?
- **Communication:** How do agents talk efficiently?
- **Incentives:** How to prevent selfish behavior?
- **Evaluation:** How to judge the team?
- **Safety:** How to prevent cascading failures?




These are the real research frontiers right now.
Solving any one of them well is worth millions.

IMG:https://www.openxcell.com/blog/multi-agent-systems/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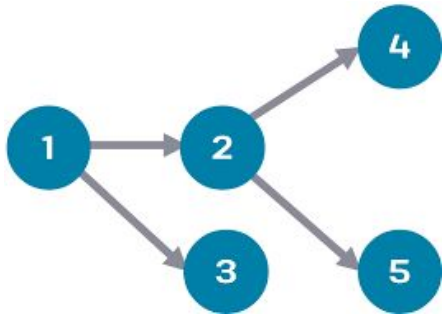
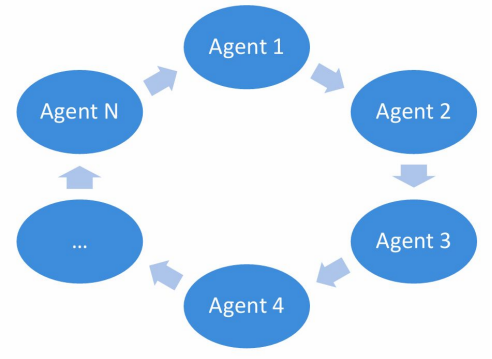
# Popular MAS Patterns - 1

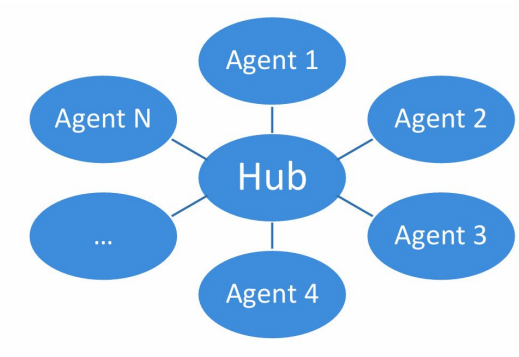
**Directed Acyclic Graph (DAG)**
Agents arranged in a dependency-based workflow; outputs flow downstream.




**Ring**
Agents communicate sequentially in a loop; ensures fair turn-taking.




**Star (Hub-and-Spoke)**
Central coordinator manages communication with all agents.

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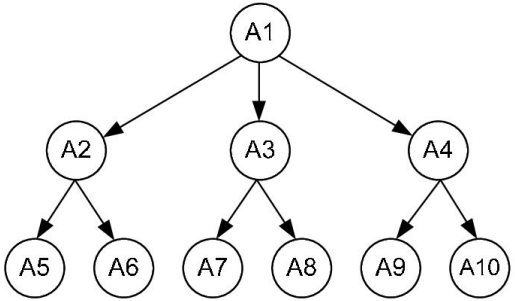
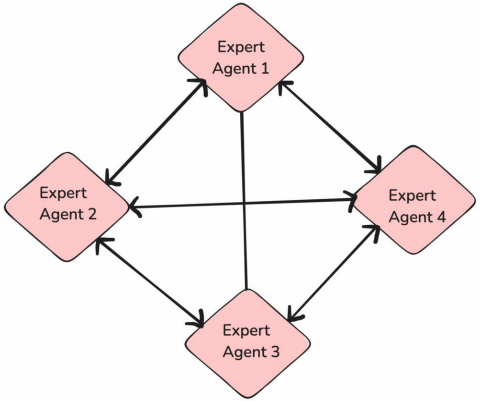
# Popular MAS Patterns - 2

**Hierarchical**
Supervisor agent decomposes tasks and delegates to worker agents.

**Conversational / Peer Coordination**
Peer agents collaborate through dialogue to reach consensus.

**Market / Auction**
Agents bid for tasks based on capability and cost; best bid wins.



IMG: https://www.couchbase.com/blog/polaris-multi-agent-conversational-ai/, https://www.dreamstime.com/illustration/auctioneer-illustration.html



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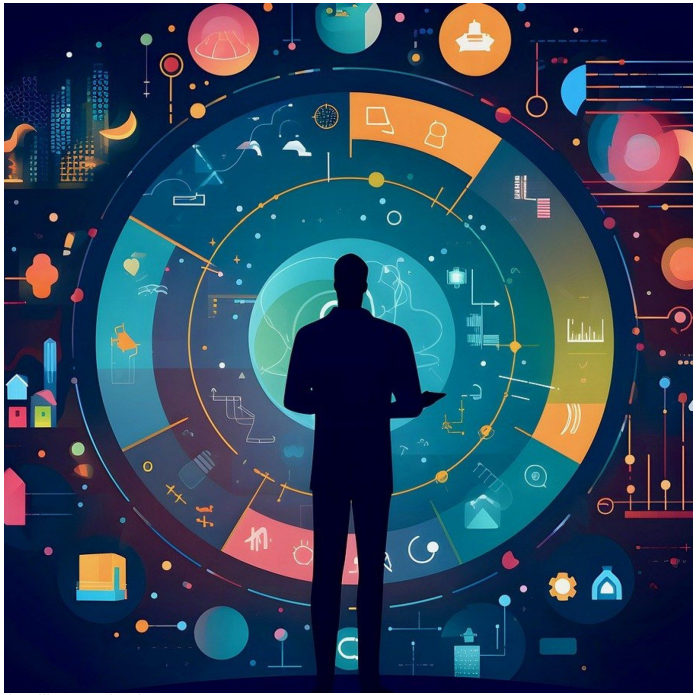
# Practical MAS Design Tips

**Recommended stack:**

- LangGraph / smolagents for orchestration
- OpenTelemetry / LangSmith for tracing & debugging
- Human-in-the-loop calibration for judges

**Key practice:**

Read traces obsessively → Tune performance budgets per agent



IMG:https://www.sciencetimes.com/articles/46966/20231107/from-data-to-visual-insights-exploring-the-crucial-role-of-artificial-intelligence-in-data-visualization.htm



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# Quick Recap : Single vs Multi-Agent

| Single Agent | Multi-Agent |
|---|---|
| PEAS → Architecture → ReAct/CoT/ToT | Teams, coordination, communication |
| Simpler, cheaper, easier to debug | Higher ceiling, more emergent power |
| Good for well-defined tasks | Better for very complex/open-ended problems |

**Rule of thumb:** start single-agent. Only go multi-agent when single agent clearly plateaus.


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# Summary & Open Floor

**Today we covered:**

- Agentic evolution & definitions
- PEAS & environment understanding
- Architectures & BDI mapping
- CoT → ReAct → ToT → Self-critique
- When & how to build multi-agent teams

Any Questions?


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